Lab 5

Part 1

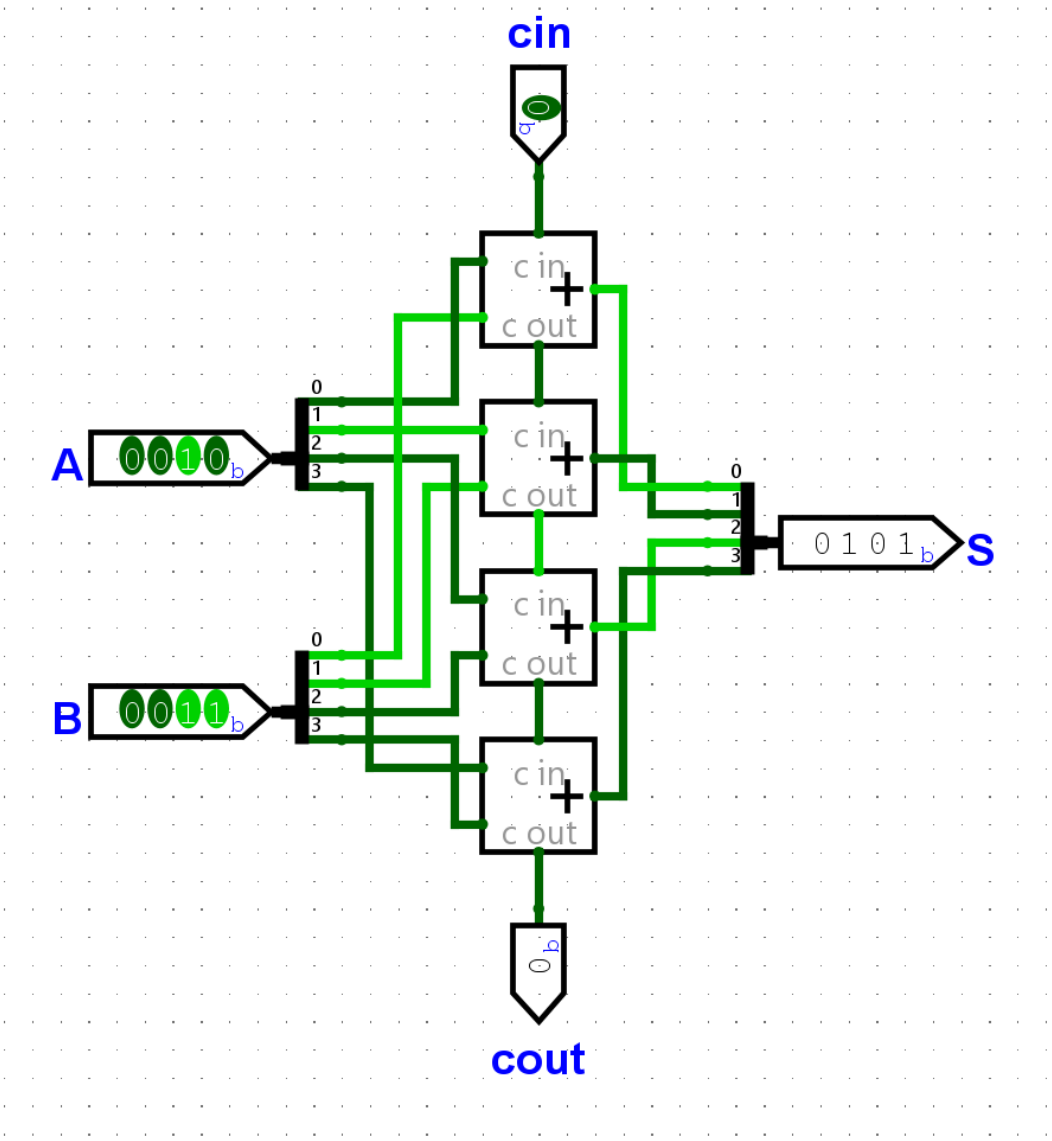


Figure 1 - 4 bit adder

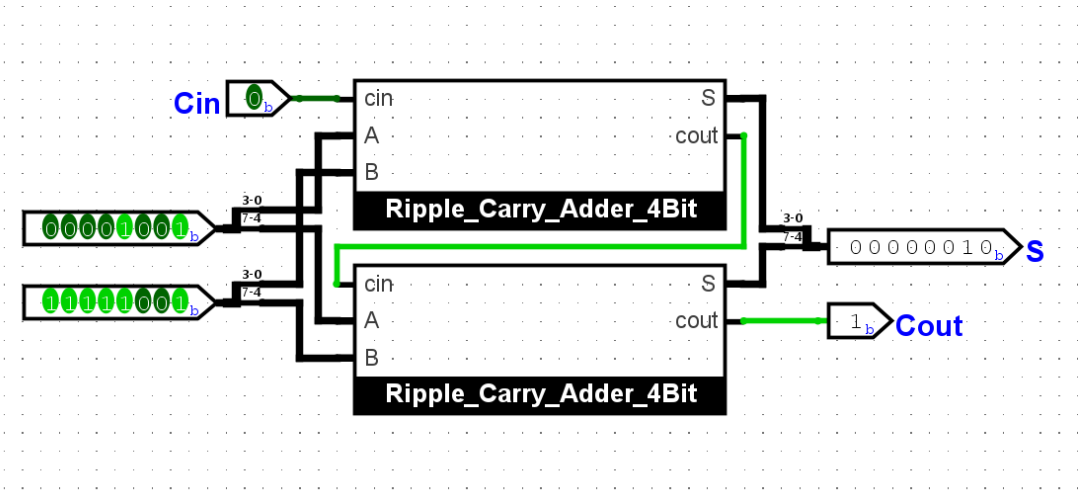


Figure 2 - Two 4bit adders to make 8bit adder

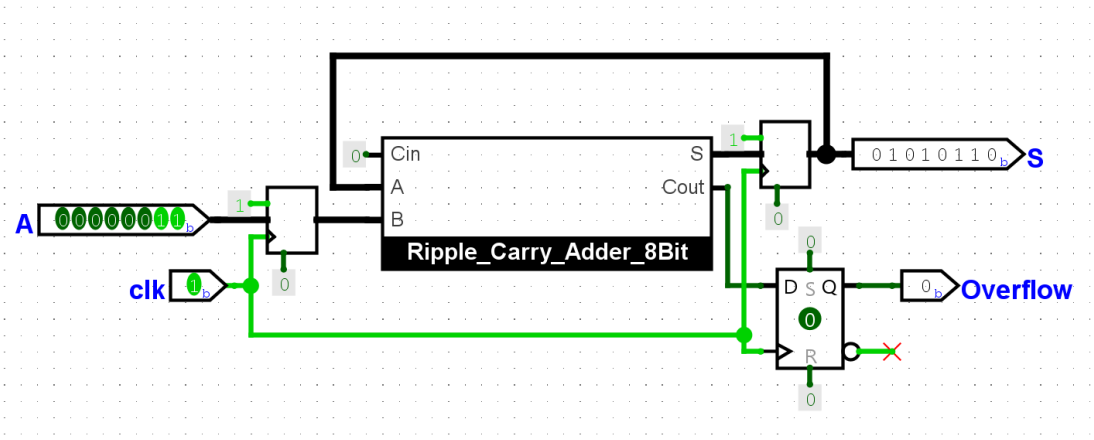


Figure 3 - The initial S state is undefined, which Logisim takes issue with. I had to dig for a certain setting that changes initial states to 0s instead.

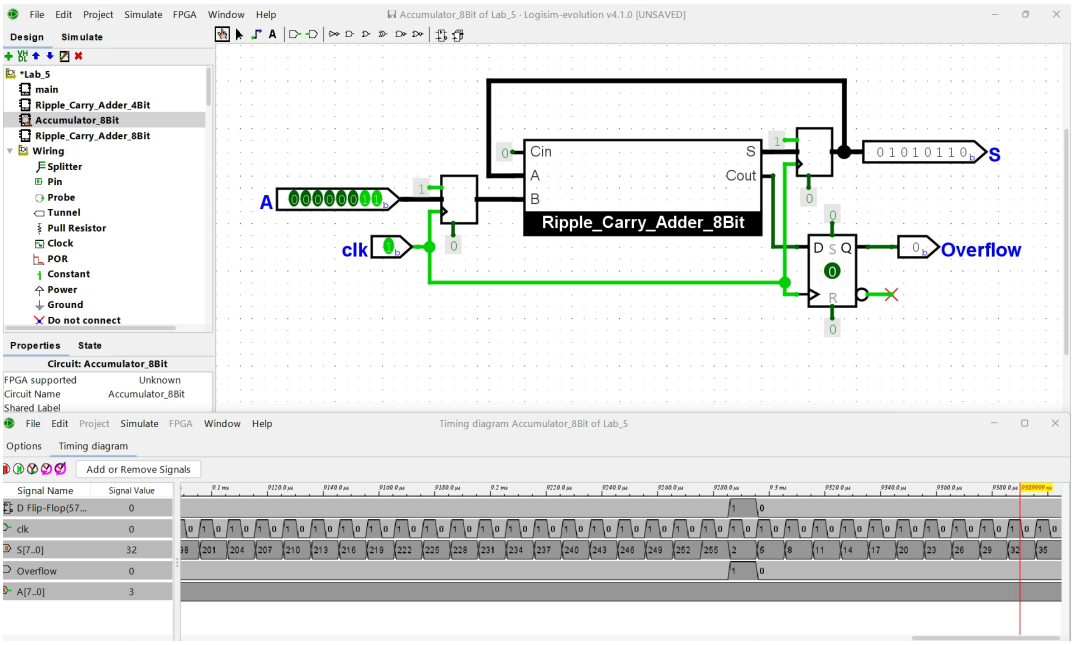


Figure 4 - Timing diagram of Accumulator

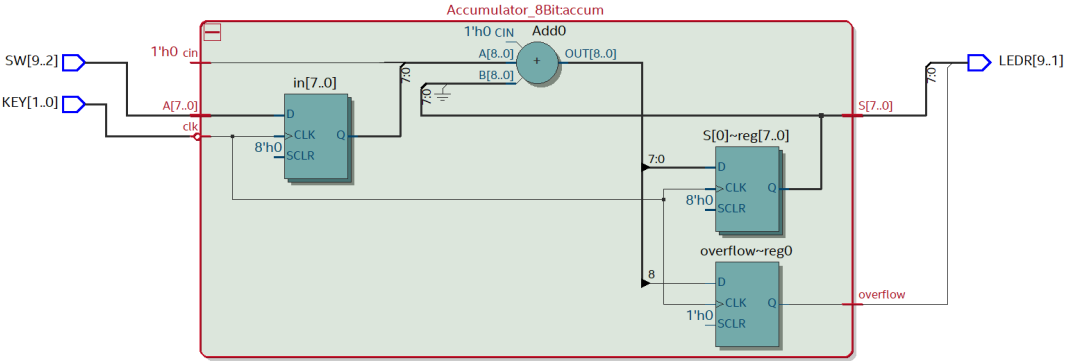


Figure 5 - Quartus RTL view of Accumulator

Part 2

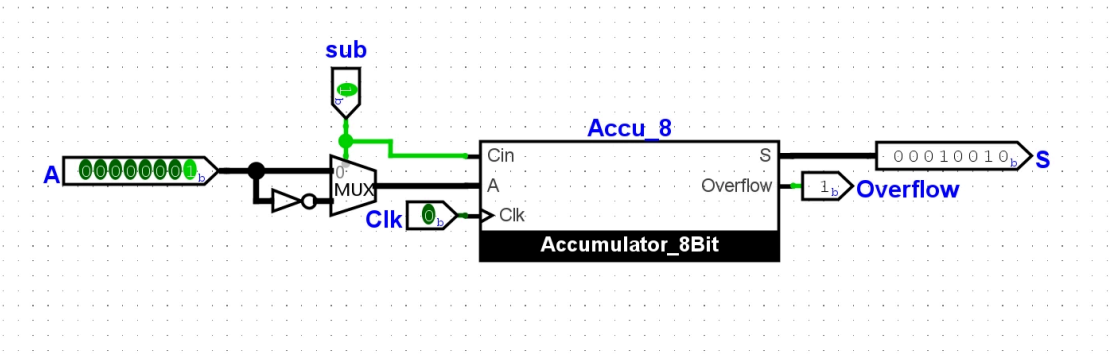


Figure 6 - Subtracting and adding accumulator with MUX

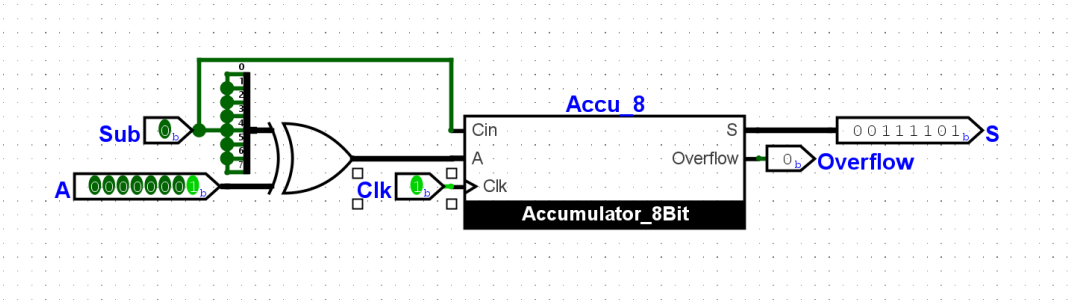


Figure 7 - Subtracting and adding accumulator with XOR

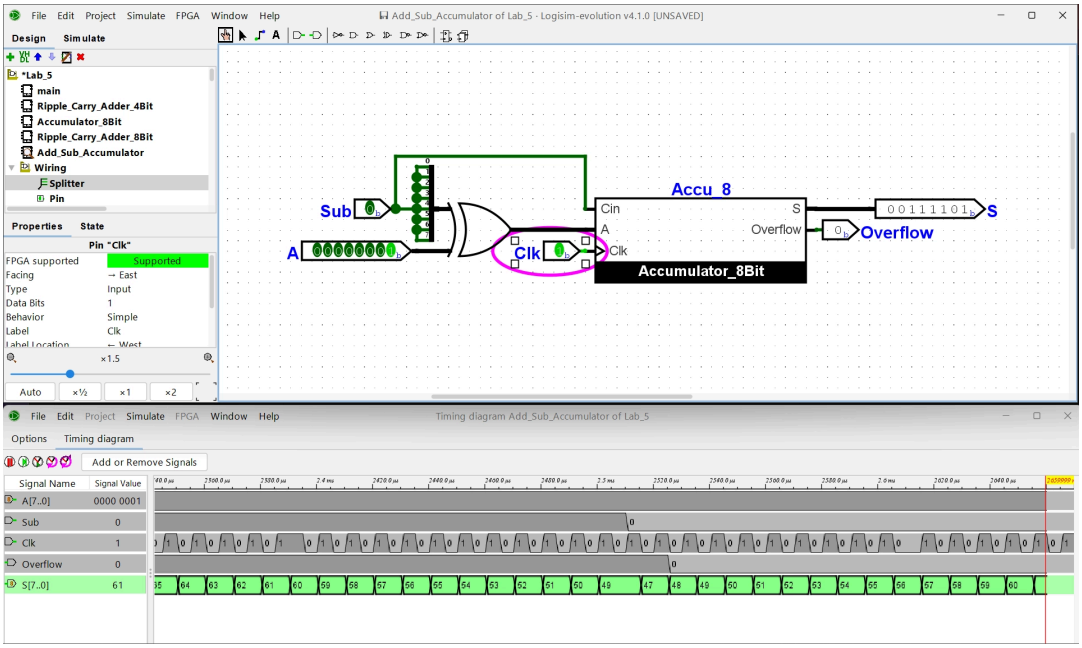


Figure 8 - When switching from subtraction to addition and vice versa, we skip a value. Not sure why yet.

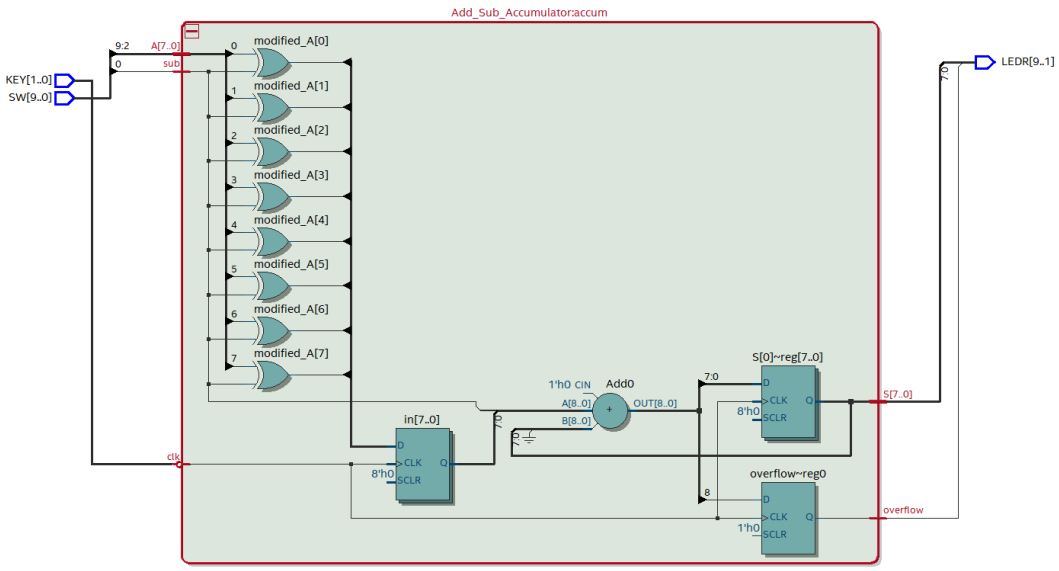


Figure 9 - RTL view showing XOR gates.

Part 3

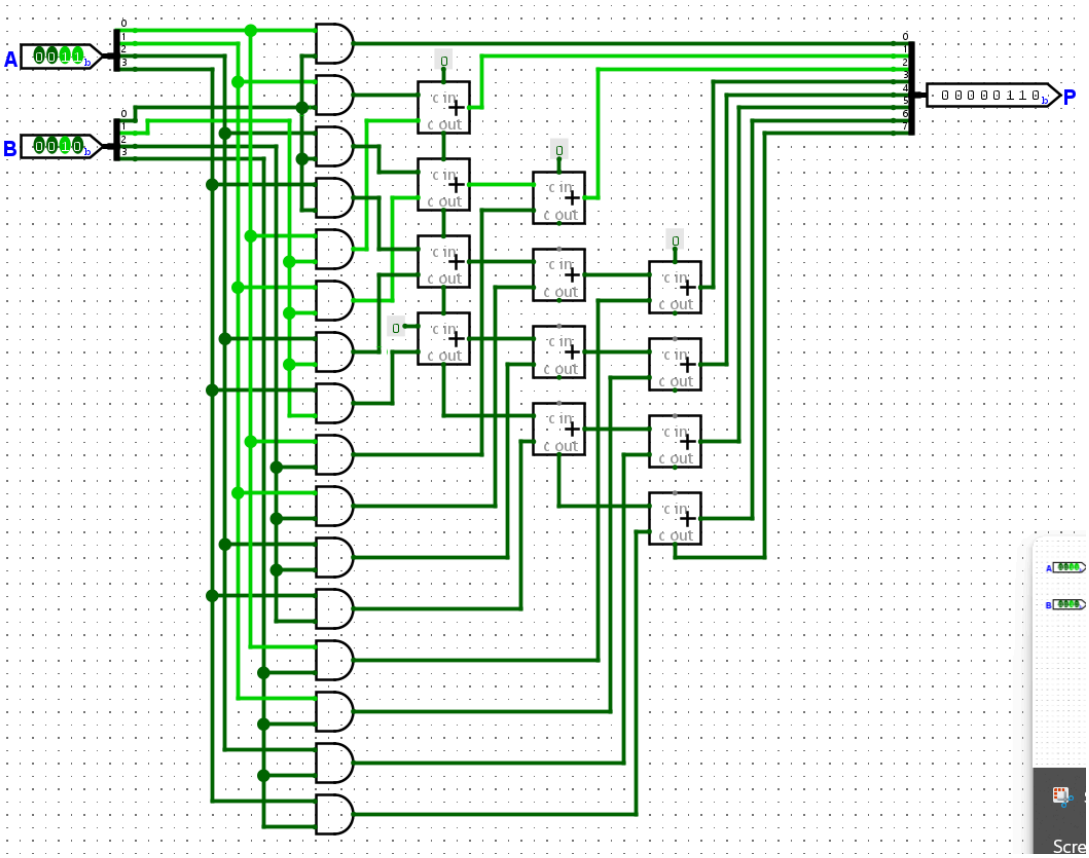


Figure 10 - 4bit multiplier circuit

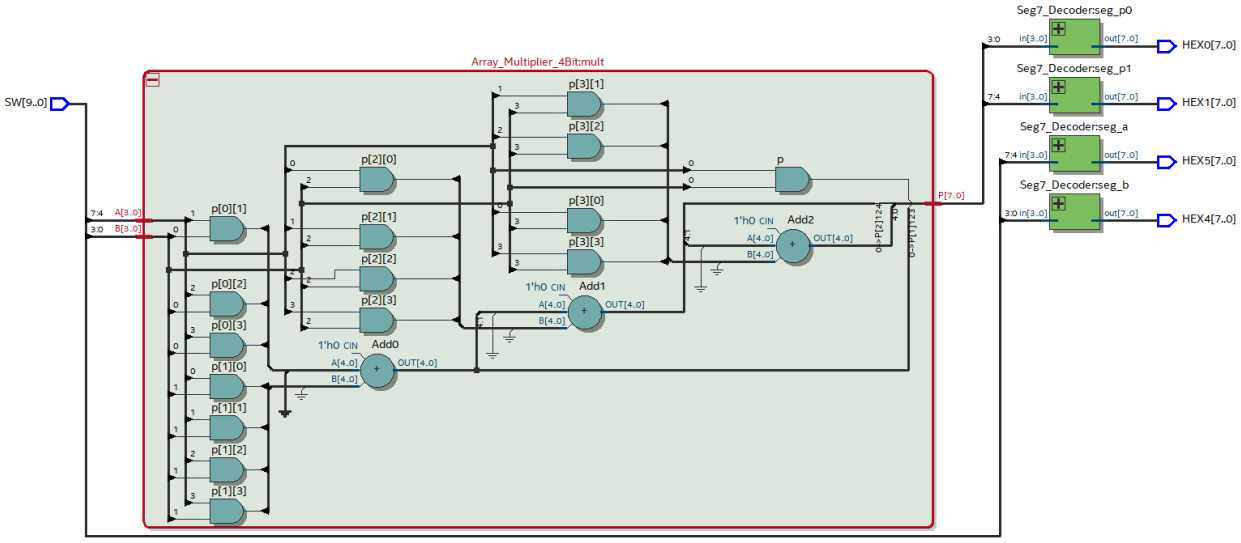


Figure 11 - 4bit multiplier circuit RTL view

Part 4

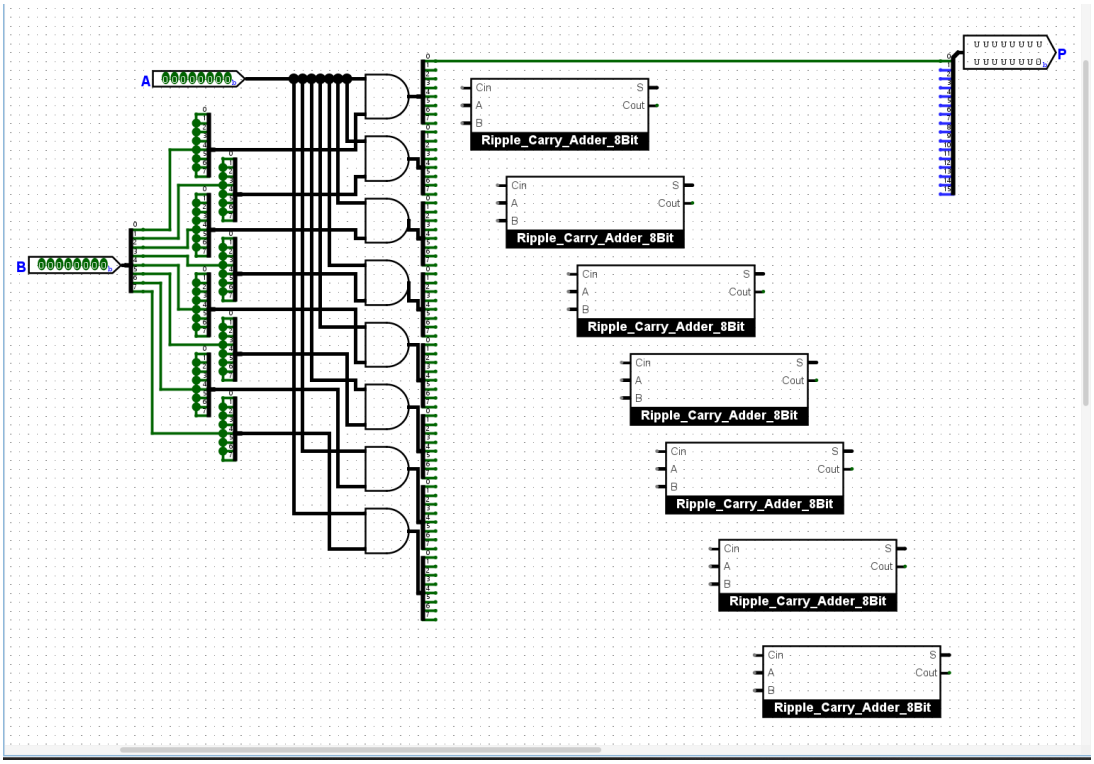


Figure 12 - Tried to make the 8×8 multiplication array, but it was taking too long and Logisim does not have the best resources. It's easy to see where this was going.

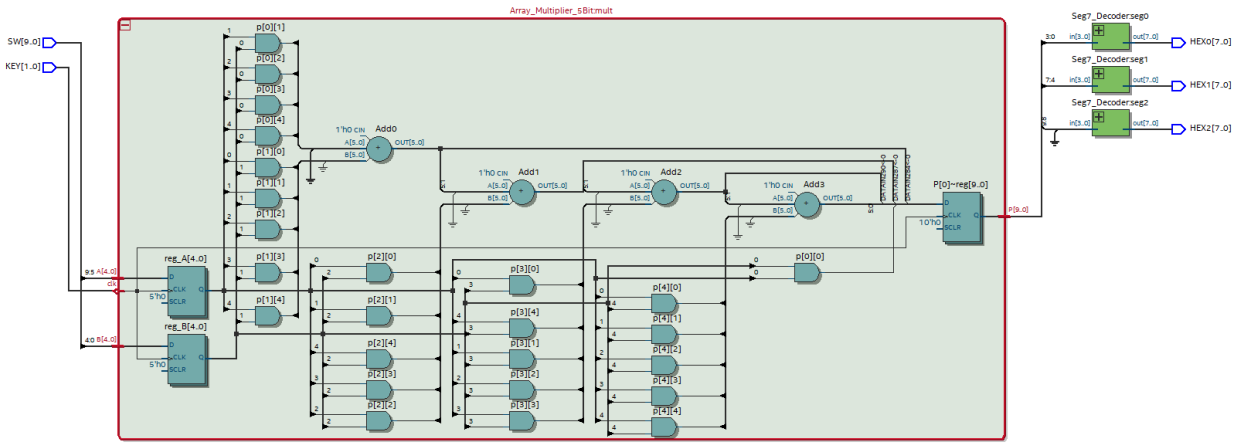


Figure 13 - 5bit multiplication circuit with registers.

Part 5